

FRC ROBOT PROJECT - FINAL FROZEN BACKBONE ARCHITECTURE

FRC Java Coding Lab 7.0 - Permanent Inheritance Development Contract

CONTROL: Driver -> Xbox Controller -> controls -> commands -> subsystems -> io -> hardware

OBSERVATION: hardware -> IOInputs -> subsystem / estimator -> immutable Observation -> telemetry -> NT4 / Glass / log

PACKAGE TREE	RESPONSIBILITY MAP	FINAL FROZEN RULES
<pre>frc.robot - Main / Robot / RobotContainer / Constants - commands/ controls/ subsystems/ - io/ `-- <mechanism>/         - <Mechanism>IO         - <Mechanism>IOReal        `-- <Mechanism>IOSim - telemetry/ - RobotTelemetry `-- <mechanism>/         - <Mechanism>TelemetryFacade        `-- focused telemetry modules `-- util/  deploy/pathplanner/  - paths/ `-- autos/</pre>	controls = Driver intent commands = Robot actions / coordination subsystems = Mechanism API and state io = Hardware contract + vendor APIs telemetry = Observation and diagnostics util = Generic shared helpers only SOURCE OF TRUTH • Hardware values originate in IO. • Mechanism state originates in subsystems/estimators. • Telemetry only publishes snapshots. • Each IO owns a mechanism-specific Inputs snapshot.	• RobotContainer: create, select, inject, configure. • Commands receive the smallest dependency needed. • Subsystems depend on IO interfaces only. • IO never publishes NetworkTables. • Telemetry never reads vendor devices or controls behavior. • Facade hides internal telemetry modules. • Constants.java is default; constants/ requires formal review. • commands/auto coordinates; PathPlanner files live in deploy/. • New mechanisms follow responsibility patterns, not file-by-file copying. • Normal lessons extend the backbone; they do not redesign it.

APPROVED FLOWS

CONTROL Driver -> Xbox Controller -> controls -> commands -> subsystems -> io -> hardware	OBSERVATION hardware -> IOInputs -> subsystem / estimator -> immutable Observation -> telemetry -> typed NT4 / Glass / log	FORBIDDEN Observation -> hardware / vendor / NT / scheduler / RobotContainer / mutable state / control Telemetry -> control IO -> NT
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STATUS: VERSION 1.1 FROZEN - APPROVED 2026-08-01

REVISION HISTORY: 1.0 (2026-07-18) Initial frozen poster | 1.1 (2026-08-01) APPROVED observation package governance migration